

COMPUTER NETWORKS

Milestone 2:

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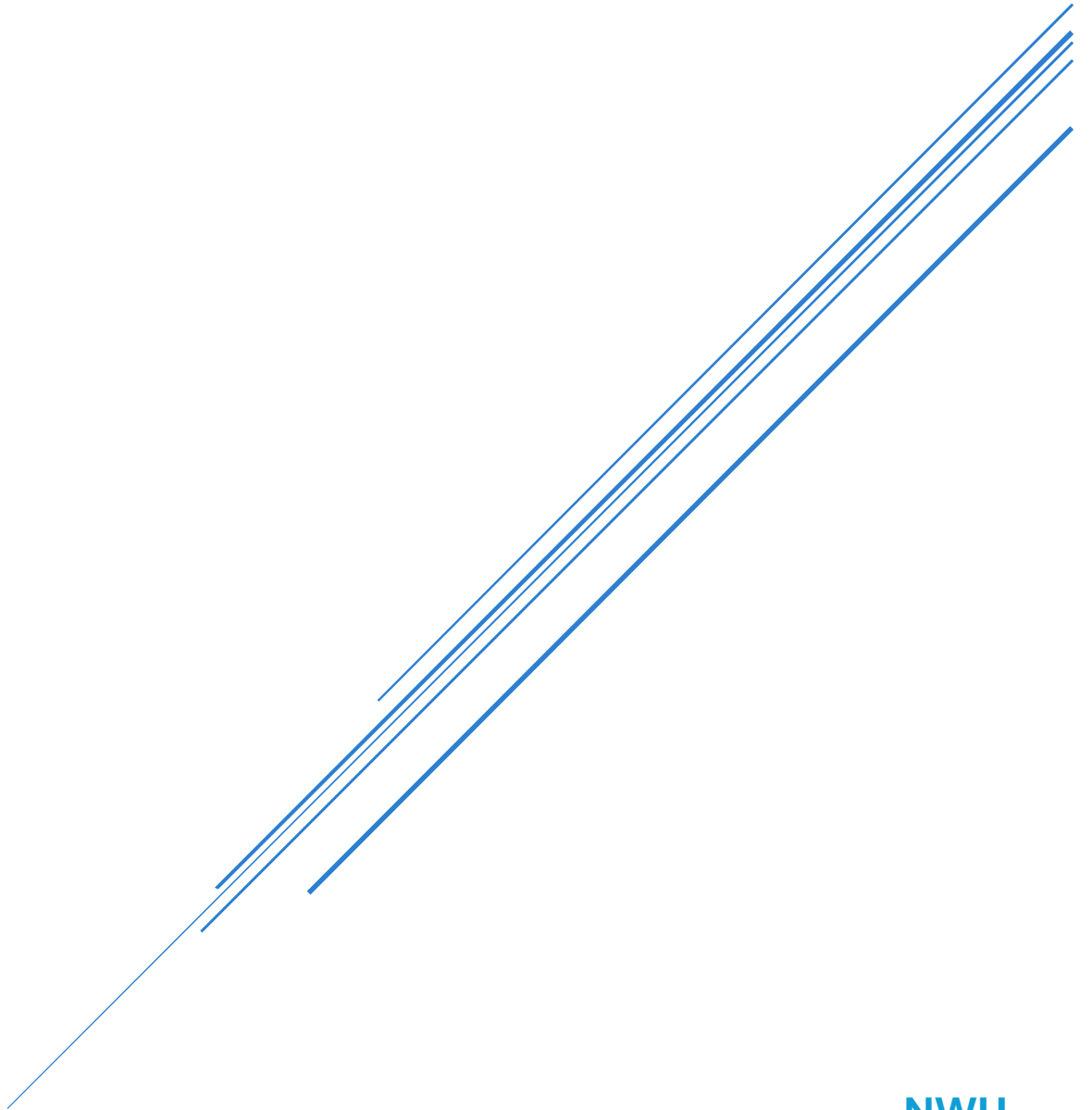


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1 Implementation Overview:

A working Cisco Packet Tracer file was developed to implement the client's requirements. The topology includes dual ISP connectivity, firewall NAT/ACL, inter-VLAN routing, and an internal HTTP server.

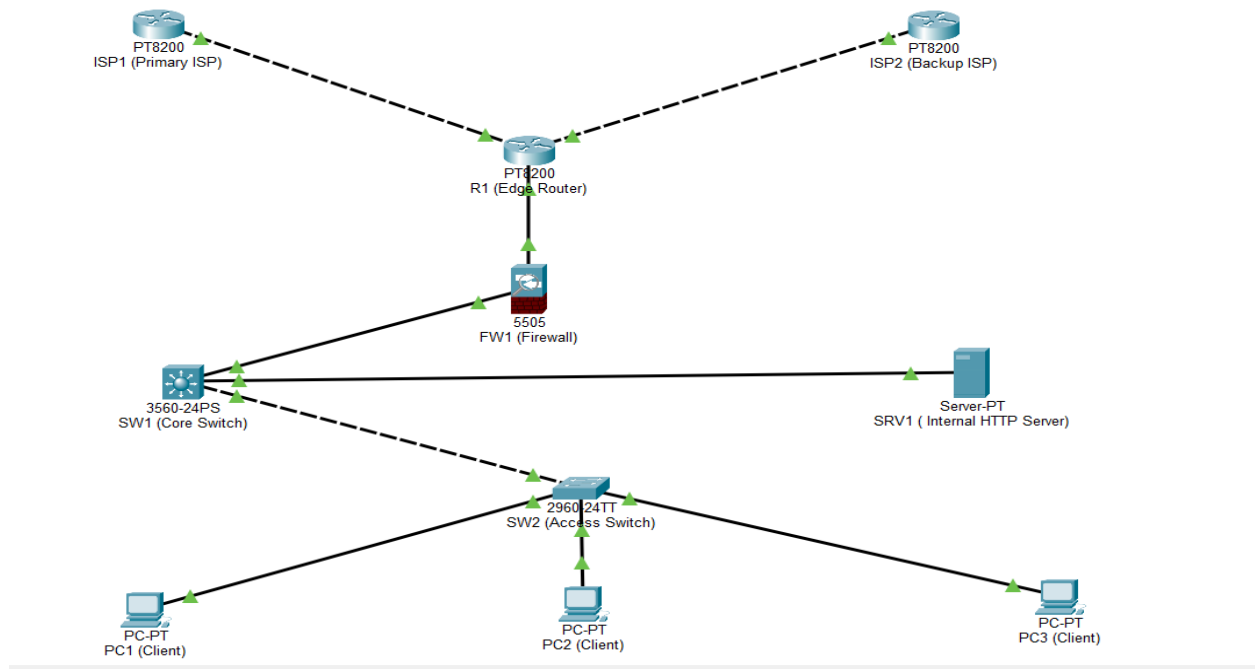


Figure 1 Full network topology diagram from Packet Tracer.

Device Roles Table:

Device	Role
R1	Dual-WAN edge router
FW1	Firewall/NAT
SW1	Layer-3 core switch
SW2	Access switch
SRV1	Internal HTTP server
PC1–PC3	Client workstations
ISP1	Primary Internet connection
ISP2	Backup Internet connection

2 Assigned Feature Overview:

Design Justification: The client required an internal web service for hosting company information. An HTTP server was chosen because it provides a simple, accessible platform for internal users without requiring external DNS or complex services. Hosting the server inside VLAN 20 (Servers VLAN) ensures segmentation and security, while inter-VLAN routing on SW1 allows client PCs in VLAN 30 to access the service.

Implementation Details:

- **Device:** SRV1 (Server-PT)
- **IP Address:** 192.168.36.18
- **Subnet Mask:** 255.255.255.240
- **Default Gateway:** 192.168.36.17
- **VLAN:** 20 (Servers VLAN)
- **Service:** HTTP enabled
- **Web Page Content:** “Tiro Mining Engineering Services”

Configuration Process:

- Enabled HTTP service in the Services tab.
- Uploaded and edited index.html using the File Manager.
- Verified service status (HTTP ON).
- Tested connectivity with pings from PCs in VLAN 30.
- Accessed the webpage via browser using `http://192.168.36.18/index.html`.

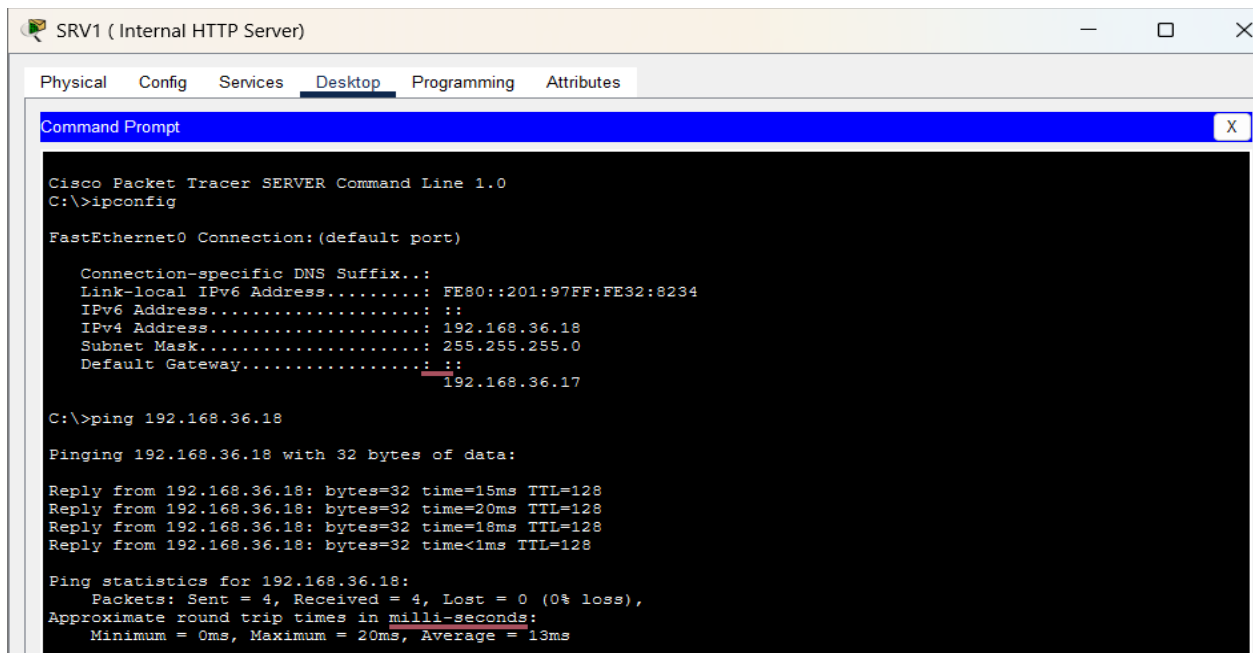
Testing Evidence:

- **Figure 3:** PC1 browser accessing SRV1 webpage.
- **Figure 4:** SRV1 Services tab showing HTTP service ON and index.html file.
- **Additional Screenshots (to add):** PC2 and PC3 browser access to SRV1 webpage, proving inter-VLAN routing.

3. Assigned Feature — Internal HTTP/Web Server

SRV1 Configuration:

Parameter	Configuration
IP Address	192.168.36.18
Subnet Mask	255.255.255.240
Default Gateway	192.168.36.17
VLAN	VLAN 20
Service	HTTP
Web Page	“Tiro Mining Engineering Services”



```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:97FF:FE32:8234
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.36.18
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                               192.168.36.17

C:\>ping 192.168.36.18

Pinging 192.168.36.18 with 32 bytes of data:

Reply from 192.168.36.18: bytes=32 time=15ms TTL=128
Reply from 192.168.36.18: bytes=32 time=20ms TTL=128
Reply from 192.168.36.18: bytes=32 time=18ms TTL=128
Reply from 192.168.36.18: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.36.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 13ms
```

Figure 2 .SRV1 IP configuration

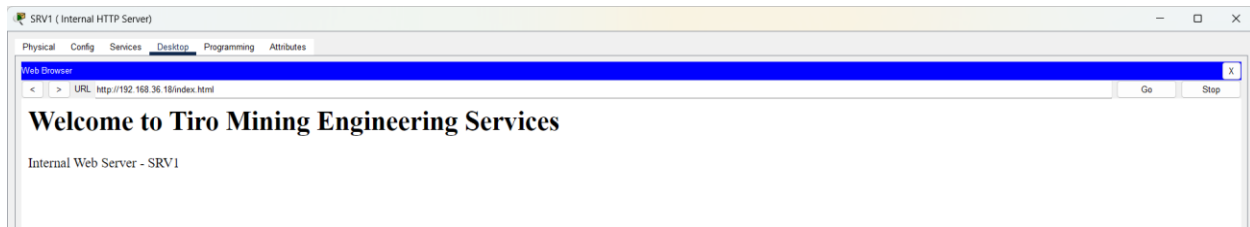


Figure 3 .PC1 browser accessing

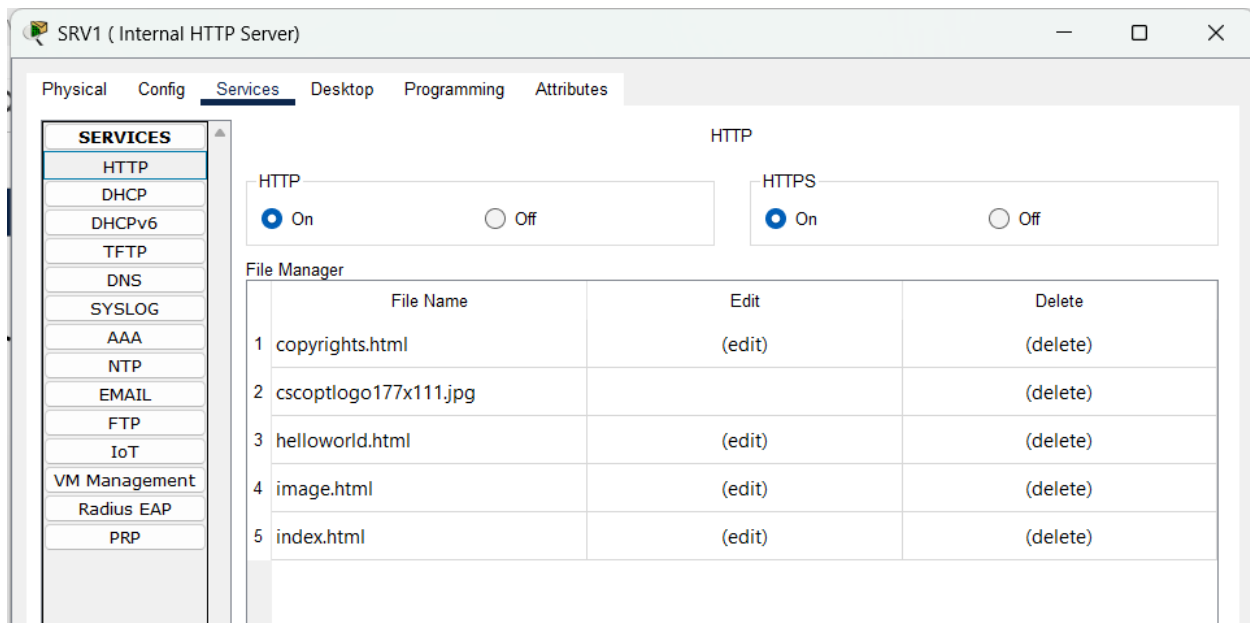


Figure 4 HTTP service enabled on SRV1.

4. Network Implementation:

4.1 R1 — Dual Internet Connectivity

- Configured ISP1 (10.0.0.2/30) and ISP2 (10.0.0.6/30).
- Primary default route via ISP1.
- Floating static backup route via ISP2.

```

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Vlan2
L       172.16.0.1/32 is directly connected, Vlan2
S       192.168.36.0/24 [1/0] via 172.16.0.2
S*      0.0.0.0/0 [1/0] via 10.0.0.1

R1#

```

Figure 5 R1 routing table showing primary ISP1 default route and floating backup ISP2 route.

4.2 FW1 — Firewall/NAT:

- Inside interface: 192.168.36.65/30.
- Outside interface: 172.16.0.2/30.
- NAT configured for inside → outside traffic.

```

FW1#show interface ip brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	unset	up	up
Ethernet0/1	unassigned	YES	unset	up	up
Ethernet0/2	unassigned	YES	unset	down	down
Ethernet0/3	unassigned	YES	unset	down	down
Ethernet0/4	unassigned	YES	unset	down	down
Ethernet0/5	unassigned	YES	unset	down	down
Ethernet0/6	unassigned	YES	unset	down	down
Ethernet0/7	unassigned	YES	unset	down	down
Vlan1	192.168.36.65	YES	manual	up	up
Vlan2	172.16.0.2	YES	manual	up	up
Vlan3	unassigned	YES	unset	up	down
Vlan10	unassigned	YES	unset	up	down

```

FW1#

```

Figure 6 FW1 interface brief showing inside/outside interfaces up/up.

```
FW1#show nat
Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic INSIDE_NET interface
   translate_hits = 161, untranslate_hits = 136

FW1#
FW1#
```

Figure 7 FW1 NAT table showing active rule and translate hits.

```
FW1#
FW1#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list INSIDE_OUT: 2 elements; name hash: 0x0cb0cb67
access-list INSIDE_OUT line 1 extended permit ip 192.168.36.0 255.255.255.0 any(hitcnt=197) 0x9cb0c425
access-list INSIDE_OUT line 2 extended permit icmp 192.168.36.0 255.255.255.0 any(hitcnt=0) 0x17268789
FW1#
```

Figure 8 FW1 ACL output showing permit rules and hit counts.

4.3 SW1 — Layer-3 Core

```
SW1>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Gig0/1, Gig0/2
10	Management	active	
20	Servers	active	Fa0/2
30	Clients	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	up	up
GigabitEthernet0/1	unassigned	YES	unset	down	down
GigabitEthernet0/2	unassigned	YES	unset	down	down
Vlan1	192.168.36.66	YES	manual	up	up
Vlan10	192.168.36.1	YES	manual	up	up
Vlan20	192.168.36.17	YES	manual	up	up
Vlan30	192.168.36.33	YES	manual	up	up

Figure 9 VLAN brief and IP interface brief showing VLANs and SVIs.


```

interface FastEthernet0/23
!
interface FastEthernet0/24
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.36.66 255.255.255.252
!
interface Vlan10
 mac-address 0001.642d.8b01
 ip address 192.168.36.1 255.255.255.240
!
interface Vlan20
 mac-address 0001.642d.8b02
 ip address 192.168.36.17 255.255.255.240
!
interface Vlan30
 mac-address 0001.642d.8b03
 ip address 192.168.36.33 255.255.255.224
!
ip default-gateway 192.168.36.1
ip classless
ip route 0.0.0.0 0.0.0.0 172.16.0.2
ip route 0.0.0.0 0.0.0.0 192.168.36.65
!
ip flow-export version 9
.

```

Figure 10b SW1 running-config showing DHCP pool definitions.

```

SW1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.36.65 to network 0.0.0.0

    192.168.36.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.36.0/28 is directly connected, Vlan10
C       192.168.36.16/28 is directly connected, Vlan20
C       192.168.36.32/27 is directly connected, Vlan30
C       192.168.36.64/30 is directly connected, Vlan1
S*    0.0.0.0/0 [1/0] via 192.168.36.65

SW1#
SW1#

```

Figure 11 SW1 routing table showing connected VLANs and default route to FW1.

4.4 SW2 — Access Layer:

```
SW2#show running-config
Building configuration...

Current configuration : 1387 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW2
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
--More--
```

SW2 (Access Switch)

Physical Config CLI Attributes

```
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan10
 ip address 192.168.36.3 255.255.255.240
!
ip default-gateway 192.168.36.1
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end
```

Figure 12 SW2 running-config showing access port VLAN assignments.

```

SW2#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    10,20,30

Port      Vlans allowed and active in management domain
Fa0/24    10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20,30
-----

```

Figure 13 SW2 trunk interface output showing VLANs allowed.

5. Testing Evidence

5.1 Internal Connectivity

PC1(Client)

```

C:\>ping 192.168.36.33

Pinging 192.168.36.33 with 32 bytes of data:

Reply from 192.168.36.33: bytes=32 time=1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.36.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

Figure 14. PC1 ping to 192.168.36.33 (gateway success).

```

C:\>ping 192.168.36.18

Pinging 192.168.36.18 with 32 bytes of data:

Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.36.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure 15 PC1 ping to 192.168.36.18 (server success).

PC2(Client)

```
C:\>ping 192.168.36.33

Pinging 192.168.36.33 with 32 bytes of data:

Request timed out.
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.36.33:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 16 PC2 ping to 192.168.36.33 (gateway success).

```
C:\>ping 192.168.36.18

Pinging 192.168.36.18 with 32 bytes of data:

Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time=19ms TTL=127
Reply from 192.168.36.18: bytes=32 time=17ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.36.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 19ms, Average = 9ms
```

Figure 17 PC2 ping to 192.168.36.18 (server success).

PC3(Client):

```
C:\>ping 192.168.36.33

Pinging 192.168.36.33 with 32 bytes of data:

Request timed out.
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255
Reply from 192.168.36.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.36.33:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 18 PC3 ping to 192.168.36.33 (gateway success).

```

C:\> ping 192.168.36.18

Pinging 192.168.36.18 with 32 bytes of data:

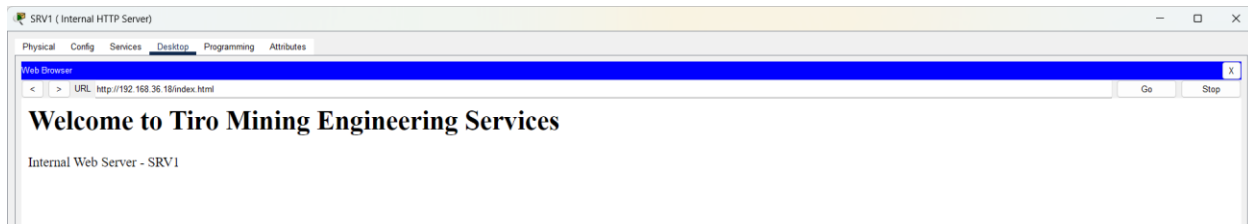
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127
Reply from 192.168.36.18: bytes=32 time=1ms TTL=127
Reply from 192.168.36.18: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.36.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

Figure 19 PC 3 ping to 192.168.36.18 (server success)

5.2 HTTP Server Testing



5.3 Firewall/NAT Testing

```

FW1#show nat
Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic INSIDE_NET interface
   translate_hits = 345, untranslate_hits = 136
FW1#

```

Figure 20 FW1 NAT table showing active translations.

```

FW1#show xlate
0 in use, 0 most used
FW1#
FW1#

```

Figure 21 FW1 NAT translation entries.

```

FW1#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list INSIDE_OUT; 2 elements; name hash: 0xe29d4436
access-list INSIDE_OUT line 1 extended permit ip 192.168.36.0 255.255.255.0 any(hitcnt=243) 0x26134964
access-list INSIDE_OUT line 2 extended permit icmp 192.168.36.0 255.255.255.0 any(hitcnt=0) 0xc034b3ad
FW1#

```

Figure 22 FW1 ACL permit rules with hit counts.

5.4 CR15 Failover Testing:

```
R1>enable
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Vlan2
L       172.16.0.1/32 is directly connected, Vlan2
S       192.168.36.0/24 [1/0] via 172.16.0.2
S*      0.0.0.0/0 [1/0] via 10.0.0.1

R1#
```

Figure 23 R1 routing table with ISP1 active.

```
R1#shutdown
Translating "shutdown"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Vlan2
L       172.16.0.1/32 is directly connected, Vlan2
S       192.168.36.0/24 [1/0] via 172.16.0.2
S*      0.0.0.0/0 [1/0] via 10.0.0.1

R1#
```

Figure 24 R1 routing table after ISP1 failure

6 Updated GitHub Portfolio

https://github.com/Samkelokhza/Computer_Networks-Tiro_Mining_Network_Project

7 .References:

1. Cisco Systems. *Cisco Packet Tracer – Network Simulation Tool User Guide*. Cisco Networking Academy, 2024.

This guide provides detailed instructions on configuring devices, enabling services such as HTTP, and simulating failover scenarios in Packet Tracer. It supports the implementation of VLANs, NAT, ACLs, and inter-VLAN routing as demonstrated in Milestone 2.

2. Tanenbaum, A. S., & Wetherall, D. J. *Computer Networks*. 5th Edition. Pearson Education, 2011.

A foundational textbook explaining client-server communication, IP addressing, routing, and firewall concepts. It underpins the theoretical justification for implementing an internal HTTP server, NAT translation, and dual-ISP failover in your project.